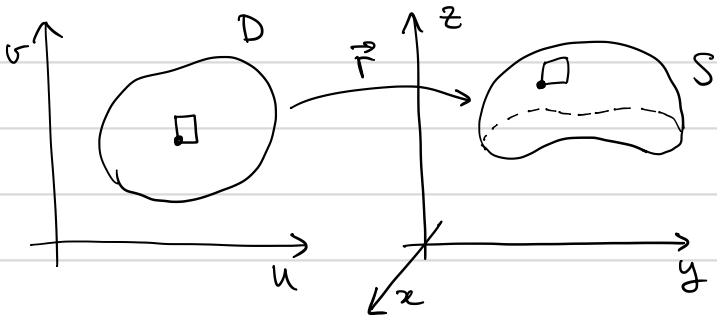
Lecture 21

surface area (16.6)

surface integrals (16.7)



$$\vec{r}(u,v) = \langle x(u,v), y(u,v), z(u,v) \rangle$$

$$\boxed{\text{Area}(S) = \iint_D |\vec{r}_u \times \vec{r}_v| \, dA}$$

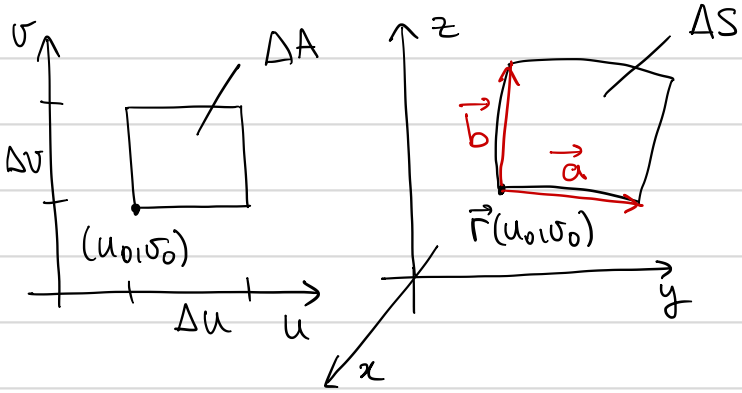
defined by Riemann sums

$$\boxed{\iint_S f(x,y,z)\, dS = \iint_D f(\vec{r}(u,v)) |\vec{r}_u \times \vec{r}_v| \, dA}$$

Interpretation:

$$\iint_S f(x,y,z)\, dS = \text{Area}(S) \cdot \left(\begin{array}{c}\text{Average value of} \\ f \text{ on } S\end{array}\right)$$

Zoom:



$$\vec{a} = \vec{r}(u_0+\Delta u, v_0) - \vec{r}(u_0,v_0)$$
$$\approx \frac{\partial \vec{r}}{\partial u}(u_0,v_0)\,\Delta u$$

$$\vec{b} = \vec{r}(u_0, v_0+\Delta v) - \vec{r}(u_0,v_0)$$
$$\approx \frac{\partial \vec{r}}{\partial v}(u_0,v_0)\,\Delta v$$

area parallelogram

$$\Delta S \approx |\vec{a} \times \vec{b}| = |\vec{r}_u \times \vec{r}_v| \, \Delta u \, \Delta v$$

Example: S = sphere of radius a

$$\vec{r}(\phi,\theta) = (a \sin\phi \cos\theta,\ a \sin\phi \sin\theta,\ a\cos\phi)$$

$$\frac{\partial \vec{r}}{\partial \phi} \times \frac{\partial \vec{r}}{\partial \theta} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial x}{\partial \phi} & \frac{\partial y}{\partial \phi} & \frac{\partial z}{\partial \phi} \\ \frac{\partial x}{\partial \theta} & \frac{\partial y}{\partial \theta} & \frac{\partial z}{\partial \theta} \end{vmatrix}$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a\cos\phi\cos\theta & a\cos\phi\sin\theta & -a\sin\phi \\ -a\sin\phi\sin\theta & a\sin\phi\cos\theta & 0 \end{vmatrix}$$

$$= a^2 \sin\phi \left(\sin\phi(\cos\theta\,\hat{i} + \sin\theta\,\hat{j}) + \cos\phi\,\hat{k}\right)$$

$$|\vec{r}_\phi \times \vec{r}_\theta| = a^2 \sin\phi \qquad (0 \le \phi \le \pi)$$

$$\text{Area}(S) = \iint_D |\vec{r}_\phi \times \vec{r}_\theta| \, dA$$

$$= \int_{\phi=0}^{\pi} \int_{\theta=0}^{2\pi} a^2 \sin\phi \, d\theta \, d\phi$$

$$= 2\pi a^2 \int_0^{\pi} \sin\phi \, d\phi = 4\pi a^2 \,//$$

<u>Example</u>: $\iint_S x^2 \, dS =$

$$= \int_{\phi=0}^{\pi} \int_{\theta=0}^{2\pi} \overbrace{(a\sin\phi\cos\theta)^2}^{x^2} \overbrace{(a^2\sin\phi)}^{dS} \, d\theta \, d\phi$$

$$= a^4 \int_0^{2\pi} \cos^2\theta \, d\theta \cdot \int_0^{\pi} \sin^3\phi \, d\phi$$

$$= \cdots = \frac{4\pi}{3} a^4 \,//$$

<u>Example</u>: S = graph of $f(x,y)$ over D

$$\vec{r}(x,y) = \langle x, y, f(x,y) \rangle$$

$$\vec{r}_x \times \vec{r}_y = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & f_x \\ 0 & 1 & f_y \end{vmatrix}$$

$$= \hat{i}(-f_x) + \hat{j}(-f_y) + \hat{k} \cdot 1$$

$$|\vec{r}_x \times \vec{r}_y| = \sqrt{(f_x)^2 + (f_y)^2 + 1}$$

$$\text{Area}(S) = \iint_D \underbrace{\sqrt{(f_x)^2 + (f_y)^2 + 1} \, dx \, dy}_{dS}$$

<u>Example</u>: S flat, contained in xy-plane:

$$\vec{r}(u,v) = \langle x(u,v), y(u,v), 0 \rangle.$$

$$\vec{r}_u \times \vec{r}_v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} & 0 \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} & 0 \end{vmatrix}$$

$$= \underbrace{\left(\frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial y}{\partial u} \frac{\partial x}{\partial v} \right)}_{\frac{\partial(x,y)}{\partial(u,v)}} \hat{k}$$

"Jacobian determinant"